

User Manual

An autonomous racing agent for
Trackmania Nations Forever



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1 Introduction

Project Apex is a self-driving race car for the game *Trackmania Nations Forever* (TMNF). It teaches itself to drive a track by practising thousands of laps and gradually getting faster, without any human showing it the way.

Everything you need is controlled from one web page in your browser, called the **Project Apex Control Dashboard**. You do not need to write any code.

1.1 What you can do

With Project Apex you can:

- **Train an agent** to drive any TMNF track, and watch it improve from a browser tab.
- **Watch the trained agent drive** a clean, fast lap inside the game.
- **Race against the agent** yourself, using its best lap as a “ghost” car on the same track.
- **Add your own tracks** and train the agent on them.

1.2 How it works

The agent looks at the game screen and a small set of numbers describing the car (speed, position on the track, and so on). Many times per second it decides which keys to press (accelerate, brake, steer left, steer right). After every lap it is rewarded for going faster and staying on the racing line, so over time it learns to drive better. You never have to label data or tell it what to do; it learns purely by practising.

1.3 How to read this manual

- Chapter 2 lists what you need before you start.
- Chapter 3 walks you through the one-time setup.
- Chapter 4 is the main chapter: how to use the dashboard day to day.
- Chapter 5 shows how to watch and race the agent.

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- Chapter 6 explains how to add your own track.
- Chapter 7 helps if something goes wrong.

2 What You Need

Before you install Project Apex, make sure you have the following. This is a one-time check.

2.1 A suitable computer

- A **Windows 11** PC (Linux also works, but this manual uses Windows).
- An **NVIDIA graphics card** (GPU) with current drivers. The GPU is what makes training fast.
- At least **20 GB of free disk space** and **20 GB of memory (RAM)**.

2.2 Software

- **Trackmania Nations Forever (TMNF)** – a free game, easiest to install from Steam. Note: this is the older, free Trackmania, not Trackmania 2020.
- **TMLoader** and **TMInterface 2.1.0** – two small helper tools that let Project Apex talk to the game. Both are free.
- **Python 3.11** – the programming language Project Apex runs on. You will not write any Python; it just needs to be installed.
- **The Project Apex files** – the [Code](#) folder from this project.

Chapter 3 explains how to install each of these in order.

2.3 Optional: a ready-trained agent

Training an agent from scratch takes several hours. If you would rather watch and race a finished agent straight away, you can download a set of ready-trained agents (the “model weights”). The download link is given in the project

[README](#). Place the downloaded

[save](#) folder inside

[Code/linesight/](#) and you are ready to go.

3 Installation

This is a one-time setup. Follow the steps in order. Allow about two hours, mostly downloads.

3.1 Step 1 – Install the game

Install **Trackmania Nations Forever** from Steam (search for it; it is free). Launch it once, let any update finish, and note your in-game nickname – you will need it later.

3.2 Step 2 – Install TMLoader and TMInterface

1. Go to <https://donadigo.com/tminterface/>.
2. Download and install **TMLoader**. It finds your game automatically.
3. From the same page, download and install **TMInterface 2.1.0**.
4. From now on, always start the game through **TMLoader**, not directly. When you launch it, a small “TMInterface” message should appear in the game.

3.3 Step 3 – Install Python 3.11

Install **Python 3.11** from <https://www.python.org>. During installation, tick the box “Add Python to PATH”.

3.4 Step 4 – Set up Project Apex

Open a terminal (PowerShell) inside the

Code folder of this project and run the following. The first command creates an isolated environment; the second installs everything Project Apex needs.

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements-dashboard.txt
cd linesight
pip install -e .
```

3.5 Step 5 – Set the game to its lowest quality

Training is faster when the game runs as fast as possible. Start the game through TM-Loader and, in the game settings, choose **windowed mode**, the **lowest resolution**, and the **lowest graphics quality**. Picture quality does not matter to the agent.

3.6 Step 6 – Tell Project Apex about your setup

The dashboard (Chapter 4) has a **Settings** tab where you enter a few details specific to your computer: where the game is installed, where TMLoader is, and your in-game nickname. You only do this once.

That is the whole setup. From here on, everything happens in the dashboard.

4 Using the Control Dashboard

The dashboard is the control centre for Project Apex. From it you start and stop training, watch progress, adjust settings, and race the agent, all from your browser.

4.1 Opening the dashboard

Open a terminal in the

[Code](#) folder, activate the environment, and start the dashboard:

```
.venv\Scripts\activate  
streamlit run dashboard.py
```

Your browser opens at <http://localhost:8501>. Figure 4.1 shows the main page.

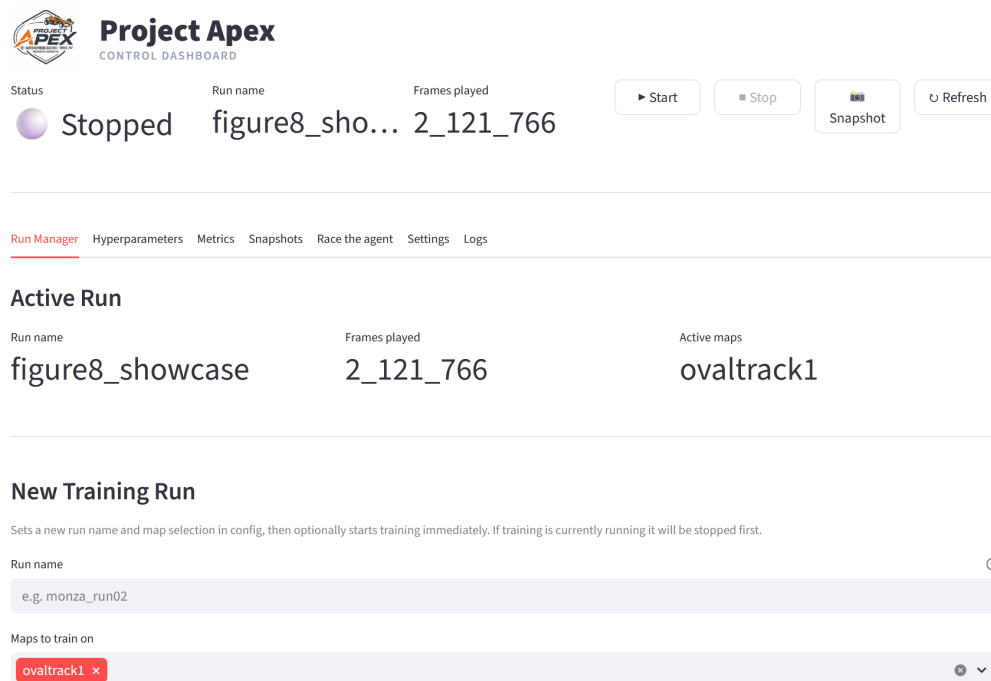


Figure 4.1: The dashboard main page. The status bar at the top shows whether training is running, the current run name, and the action buttons.

4.2 The status bar

The bar across the top is always visible. It shows:

- a **Running** or **Stopped** badge,
- the name of the current training run,
- how many practice steps the agent has done so far, and
- the buttons **Start**, **Stop**, **Snapshot**, and **Refresh**.

4.3 The tabs

Below the status bar are several tabs. You will mostly use the first few.

4.3.1 Settings – do this first

The **Settings** tab is where you enter the details for your computer: the game folder, the TMLoader program, your in-game nickname, and the connection port. Fill these in once (Figure 4.2).

4 Using the Control Dashboard

Run Manager Hyperparameters Metrics Snapshots Race the agent **Settings** Logs

Machine Configuration

These values are stored in `linesight/config_files/user_config.py`. Changing them here updates that file directly — no Python editing needed. Changes take effect the next time training starts.

TrackMania username

The in-game profile name you use in TMNF. Used for hiding personal-best replays during training/playback.

labHSE Apply

TMLoader path (Windows only)

Path to `TMLoader.exe`. On a Steam install this is usually inside `C:\Program Files (x86)\Steam\steamapps\common\TrackMania Nations Forever\`. If you installed via Epic Games or a custom location, update it here.

C:\Program Files (x86)\Steam\steamapps\common\TrackMania Nations Forever\TMLoader.exe Found Apply

TrackMania base path

Root of your TrackMania data folder. Default: `C:\Users\YOU\Documents\TrackMania`. Used to locate challenge files, replays, and autosaves.

C:\Users\Wings\Documents\TrackMania Found Apply

TMInterface port

Port number for the first TMInterface instance. Default: 8478. Only change if that port is in use.

8478 - + Apply

Figure 4.2: The Settings tab. Enter your game paths and nickname here once.

4.3.2 Run Manager – pick what to train

A “run” is one training session for one agent. The **Run Manager** tab lets you create a new run, give it a name, switch between runs, or delete one. Create a run before you start training.

4.3.3 Hyperparameters – the training settings

The **Hyperparameters** tab (Figure 4.3) holds the settings that control how the agent learns and which track it drives. The defaults work well, so you can leave them alone at first. The most useful controls are:

- **Map cycle** – choose which track(s) the agent trains on.
- **Game speed** – how fast the game runs while training. A higher value trains faster; a low value gives slow-motion you can record.

Changes take effect within a second or two, without stopping training.

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[Run Manager](#) [Hyperparameters](#) [Metrics](#) [Snapshots](#) [Race the agent](#) [Settings](#) [Logs](#)

Training is stopped. Changes write to `config.py` only and apply on next Start.

★ Defaults frozen on 2026-05-22 17:38 → [linesight/config_files/config.default.py](#) [Reset to defaults](#)

Schedules — current value + flatten-from-now

Learning rate (`lr_schedule`)

7.08e-5

- +

Flatten from now

Current at step 2_121_766: 7.08e-05

Epsilon (greedy) (`epsilon_schedule`)

0.0352

- +

Flatten from now

Current at step 2_121_766: 0.0352

Epsilon-Boltzmann (`epsilon_boltzmann_schedule`)

0.0362

- +

Flatten from now

Current at step 2_121_766: 0.0362

Gamma (discount) (`gamma_schedule`)

0.99962

- +

Flatten from now

Current at step 2_121_766: 0.99962

Reward weights

Constant reward per ms (time penalty)
(`constant_reward_per_ms`)

-0.001200

- +

Apply

Current: -0.001200

Reward per metre advanced
(`reward_per_m_advanced_along_centerline`)

0.010000

- +

Apply

Current: 0.010000

Engineered reward bonuses — flat scalar at step 0 (rewrites schedule).

Speedslide

0.0000

- +

Apply Speedslide

Neoslide

0.0000

- +

Apply Neoslide

Kamikaze

0.0000

- +

Apply Kamikaze

Close-to-VCP

0.0000

- +

Apply Close-to-VCP

Figure 4.3: The Hyperparameters tab. Defaults are fine to start; the map cycle and game speed are the most useful controls.

4.3.4 Metrics — watch it learn

The **Metrics** tab shows live charts that update every few seconds (Figure 4.4). The two to watch are:

- **Zones reached** — how far around the track the agent gets. It should climb until it reaches the full lap.
- **Lap time** — how long a lap takes. Lower is better; it should fall over time.

4 Using the Control Dashboard

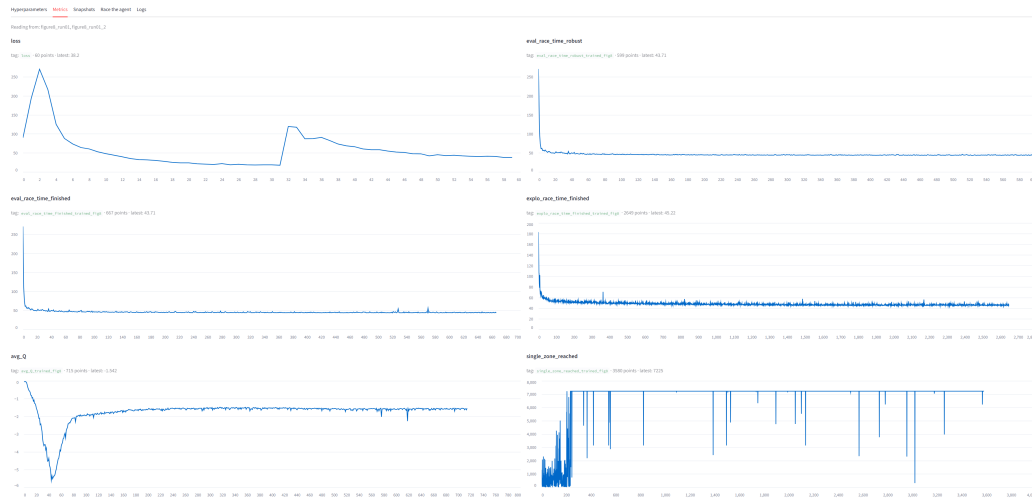


Figure 4.4: The Metrics tab. “Zones reached” climbing and “lap time” falling means the agent is learning.

4.3.5 Snapshots – save a good agent

Training keeps overwriting the agent as it goes. A **snapshot** freezes a copy of the current agent so a later, worse version cannot replace it. Use the **Snapshots** tab (or the Snapshot button) whenever the agent is driving well. The dashboard can also snapshot and stop automatically when the agent stops improving.

4.4 Starting and stopping training

1. Start the game through TMLoader, load the track, and leave the car at the start line.
2. In the dashboard, create or select a run, then press **Start**.
3. Watch the **Metrics** tab. Within about half an hour the agent should start reaching the first checkpoints; over several hours it learns to finish and then to go faster.
4. Press **Stop** when you are happy, or let the automatic stop trigger.

A rough guide: a simple oval track is usually good after a few hours; a complex circuit can take a day or more.

5 Watch and Race the Agent

Once an agent has learned a track (or you have downloaded a ready-trained one), you can watch it drive and even race against it.

5.1 Watch the agent drive

Every time the agent beats its own record during training, Project Apex saves that lap. The **Race the Agent** tab lists these best laps, fastest first (Figure 5.1).

1. Start the game through TMLoader and load the matching track.
2. In the **Race the Agent** tab, find the lap you want.
3. Press **Playback**.

The agent now drives that exact lap inside the game while you watch (Figure 5.2).

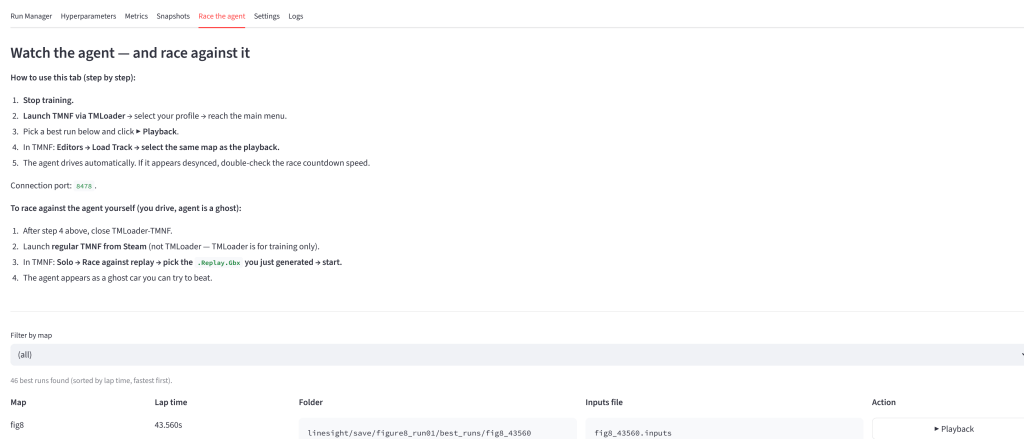


Figure 5.1: The Race the Agent tab. Best laps are listed per track, fastest first; press Playback to watch one.

5 Watch and Race the Agent

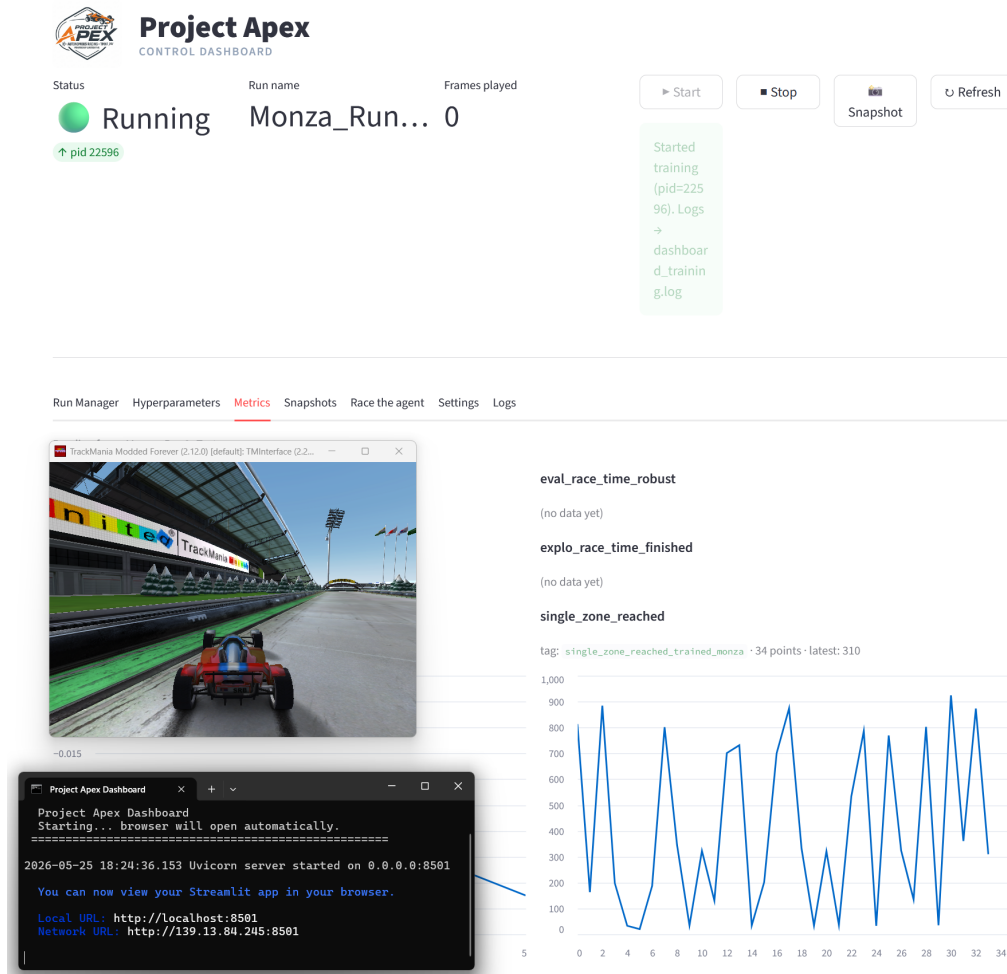


Figure 5.2: The agent driving a lap inside Trackmania Nations Forever.

5.2 Race against the agent

When a playback finishes, the game saves that lap as a normal Trackmania replay file (a “ghost”). You can then race against it:

1. Close the TMLoader version of the game and open Trackmania normally.
2. Load the same track and load the agent’s saved replay as a **ghost**.
3. Drive the track yourself. The agent appears as a see-through car running its best lap next to you.

This is the most fun way to see how good the agent has become: you are racing side by side against a driver that taught itself the track.

5.3 Tip: slow-motion for recording

If you want to record a smooth video of the agent, set the **Game speed** on the Hyperparameters tab to a low value before playback. The lap then plays in slow motion, which looks good on camera.

6 Adding Your Own Track

Project Apex can learn any TMNF track, not just the ones it comes with. Adding a track takes a few minutes plus one human lap.

1. **Get the track.** Build a track in the in-game editor, or download one. Copy its map file (ending in `.Challenge.Gbx`) into the `Code/maps` folder.
2. **Drive one reference lap.** Start the game through TMLoader, load your track, and drive a single clean lap from start to finish. Save the replay when the game offers to. This lap shows the agent roughly where the track goes; it does not need to be fast.
3. **Convert the lap.** Project Apex turns your saved replay into a guide line the agent can use. The exact command is in the project guide `docs/ADD_NEW_TRACK.md`; it is a single line you run in the terminal.
4. **Select the track.** Open the dashboard, go to the **Hyperparameters** tab, and choose your new track in the map cycle.
5. **Train as usual.** Press **Start** and watch it learn, exactly as in Chapter 4.

Simple tracks (ovals, gentle curves) learn quickly. Tracks with jumps, tight hairpins, or obstacles take longer, so give them more time.

7 Troubleshooting

If something does not work, check the table below first. Most problems come from the game not being started through TMLoader, or from a detail in the Settings tab.

Problem	What to try
The dashboard does not open in the browser	Make sure the environment is active and you ran <code>streamlit run dashboard.py</code> from the Code folder. Then open <code>http://localhost:8501</code> manually.
Training will not start, or nothing happens	Start the game through TMLoader (not directly), load the track, and leave the car at the start line before pressing Start .
The dashboard cannot find the game	Open the Settings tab and check the game folder, the TMLoader path, and your in-game nickname. The nickname must match exactly.
Training is very slow	Lower the game's resolution and graphics quality (Chapter 3, Step 5), and increase Game speed on the Hyperparameters tab.
The agent only scrapes along the wall	Give it more training time. If it still will not improve, the track's reference lap may be poor; drive a cleaner reference lap and add the track again (Chapter 6).
Playback does nothing	The game must be open through TMLoader, with the matching track loaded, before you press Playback .
A good agent was lost after more training	Use Snapshots (Chapter 4) to freeze a copy whenever the agent drives well, so later training cannot overwrite it.

Table 7.1: Common problems and how to fix them.

For anything not listed here, the detailed guides in the project

[Code/docs](#) folder (for example

[INSTALL.md](#) and

[DASHBOARD.md](#)) cover each step in more depth.